Why study the cosmic web around clusters ?

- Cluster formation and evolution is a dynamical process

Relation between X-ray derived gas properties (e.g. temperature) and dark matter properties (halo mass) should depend on accretion history

Accretion history impacts cluster galaxy properties (morphology, star-formation rate)

→ Study cosmic web filaments around clusters

- Galaxies live in groups and filaments before they enter the cluster.

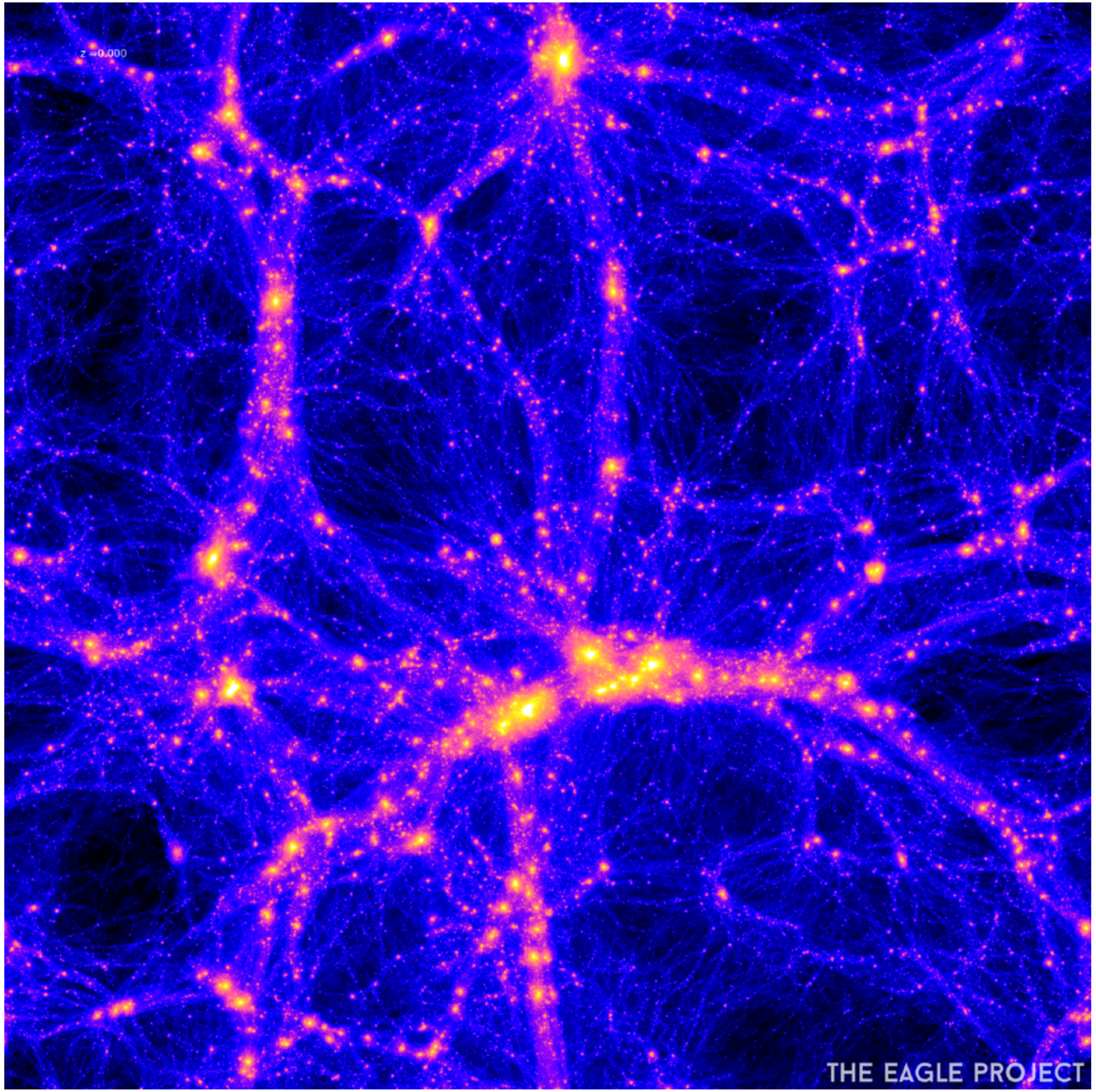
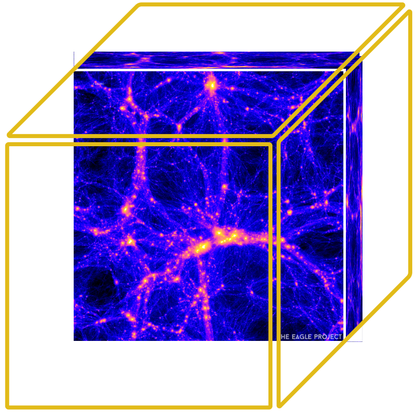
These environments influence galaxy properties

We have poor observational constraints on galaxy properties in these environments as a function of redshift

→ Cosmic groups and filaments back in time *i.e* at higher redshift

This can be done using large galaxy surveys

From dark matter to stars …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